

Trends and Developments in Aircraft Cockpits

Today's fully computerised cockpits dramatically reduce the workload of the pilot. Take a look at the new features in cockpit design that are helping to achieve this goal



Business jet—Cessna 525 CitationJet (G-SFCJ) (Courtesy: Wikipedia)

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There are some interesting developments happening in the field of aviation. Flying cars of different configurations, currently in limited production, are expected to save billions of litres of fossil fuels now wasted in traffic jams in metros all over the world. Like highways, now skyways are also being planned. There are many new entrants in the business jets manufacturing sector. Production of medium- and long-range jets has been doubled to cater to the increasing air traffic all over the world.

Integrated avionics software and large displays are creating cockpits that offer pilots dramatic new perspectives. They include QWERTY keyboards and trackballs, interfacing with a graphical 'point-and-click' display navigation system.

A conventional cockpit relies on numerous round dial mechanical gauges to display information. Today, glass cockpits look and behave like computers with Windows data that can be manipulated with mouse and other click devices—familiarily known as WIMP (Windows, Icons, Menus, Pointers) technology. Touchscreens in the cockpits are enhancing the entry and observation of flight data more efficiently.

With the increasing processing power of on-board computers, additional features like electronic flight bags (an electronic information management device) and electronic fly bags (device storing engineering and commercial data) are being incorporated in cockpits, which drastically reduce the enormous documentation.

Let's take a detailed look at these and other advancements in cockpit design.

Digital data buses

Today, like the majority of desktop computers, most operational and flying cockpits have human-machine interfaces, that are based on WIMP interfaces. Some of them standardise upon the ARINC 661 international standard digital data bus for interactivity management.

Digital data buses greatly improve the intercommunication between aircraft systems and cockpits. Previously, large amounts of aircraft wiring were required to interconnect each signal with the other equipment. As systems became more complex and more integrated, this problem aggravated. Digital data transmission techniques use links that send streams of digital data between equipment. These data links comprise only two or four twisted wires and therefore the interconnecting wiring is greatly reduced.

Common types of digital data transmission are single-source single-sink, single-source multiple-sink, multiple-source multiple-sink and fibre-optic data buses.

Single-source, multiple-sink. In this technique, one transmitting equipment can send data to a number of recipient equipment (sinks). ARINC 429 is an example of this data bus which is widely used by civil transports and business jets.

Multiple-source, multiple-sink. In this system, multiple transmitting sources may transmit data to multiple receivers. This is known as a full-duplex system and is widely employed by military users (MIL-STD-1553B) and the Boeing 777 (ARINC 629).

Major digital data buses in use